Mission 04 — Portal simulator test

Course portal · Week 4 · 3. Simulation · 'Direct Preference Optimization & Activation Steering Lab' (dlearn-cai.web.app) · tested 2026-11-01 by Squadron Shannon. Raw data: `docs/portal_simulator/portal_results.json`; reproduced with `docs/portal_simulator/verify_portal.py`. **All numeric values match.**

DPO lab, beta = 0.10 (log pi_w -12.4, log pi_l -18.2, log ref_w -15.1, log ref_l -14.8) → `DPOLoss`

quantity	portal shows	our code	match
p_win	0.648	0.6479	yes
margin	0.61	0.6100	yes
loss	0.434	0.4340	yes
grad_scale	0.0352	0.0352	yes
r_w	0.27	0.2700	yes
r_l	-0.34	-0.3400	yes

DPO lab, beta = 0.14 (log pi_w -12.4, log pi_l -18.2, log ref_w -15.1, log ref_l -14.8) → `DPOLoss`

quantity	portal shows	our code	match
p_win	0.701	0.7014	yes
margin	0.854	0.8540	yes
loss	0.3547	0.3547	yes
grad_scale	0.0418	0.0418	yes
r_w	0.378	0.3780	yes
r_l	-0.476	-0.4760	yes

SimPO lab, gamma = 0.8, beta = 2.0, |y_w| = 64, |y_l| = 240, avg log-probs -0.18 / -0.32 → `SimPOLoss`

quantity	portal shows	our code	match
sum_w	-11.5	-11.5200	yes
sum_l	-76.8	-76.8000	yes
raw_difference	65.3	65.2800	yes
r_w	-0.36	-0.3600	yes
r_l	-0.64	-0.6400	yes
margin_gap	-0.52	-0.5200	yes
p_win	0.373	0.3729	yes
loss	0.9866	0.9866	yes

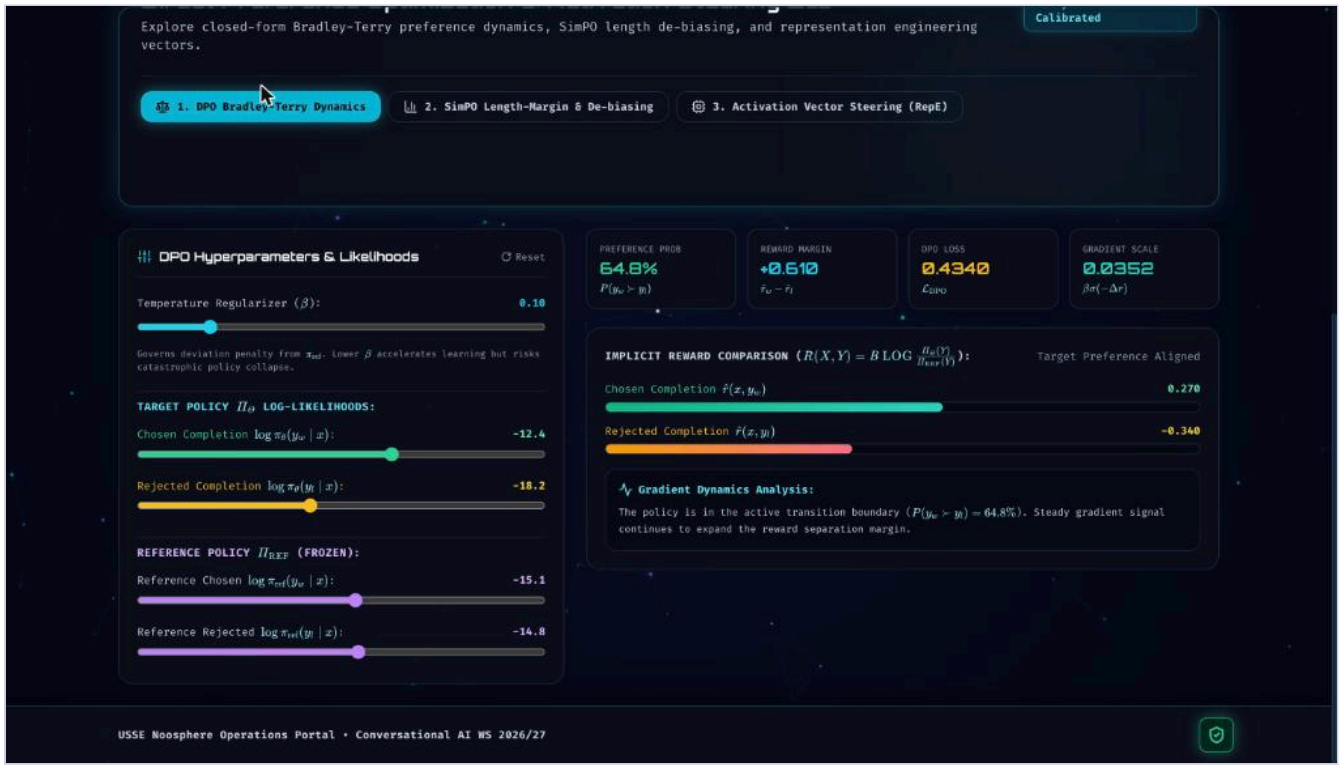
With gamma = 0.80 the portal's own example gives P = 37 %: the concise answer IS better per token (reward gap 0.28), but the gap is smaller than the target margin, so SimPO keeps pushing.

Activation steering lab

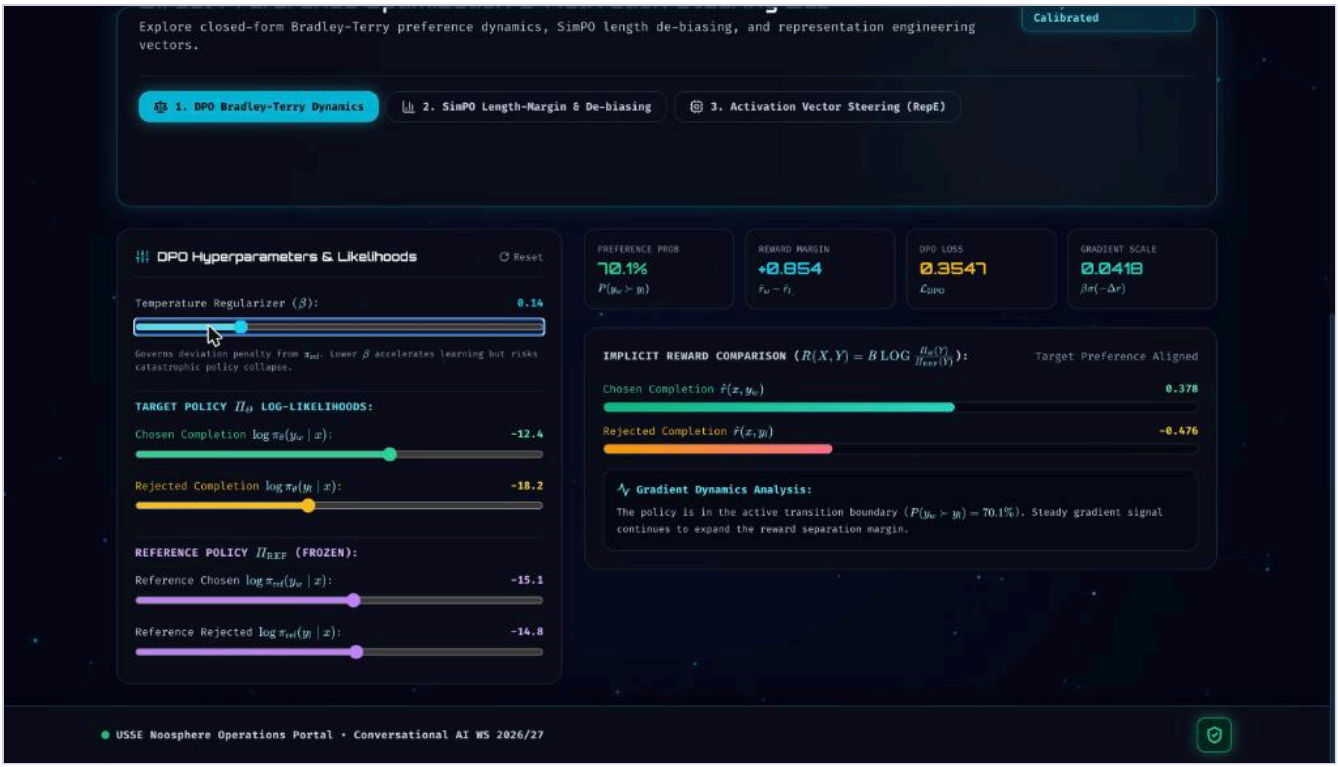
Portal claims: Layer 16 of 32 (middle layers 14-22 work best); zero additional floating-point operations during decoding.

Our measurement (Part 4, Qwen2.5-0.5B-Instruct): best layer 16 of 24 (67% depth; portal 16 of 32 = 50 %, range 44-69 %); hook cost 896 additions (one vector add) vs ~9.9e+08 FLOPs per token forward = 9.1e-07 of the cost: not zero, but negligible.

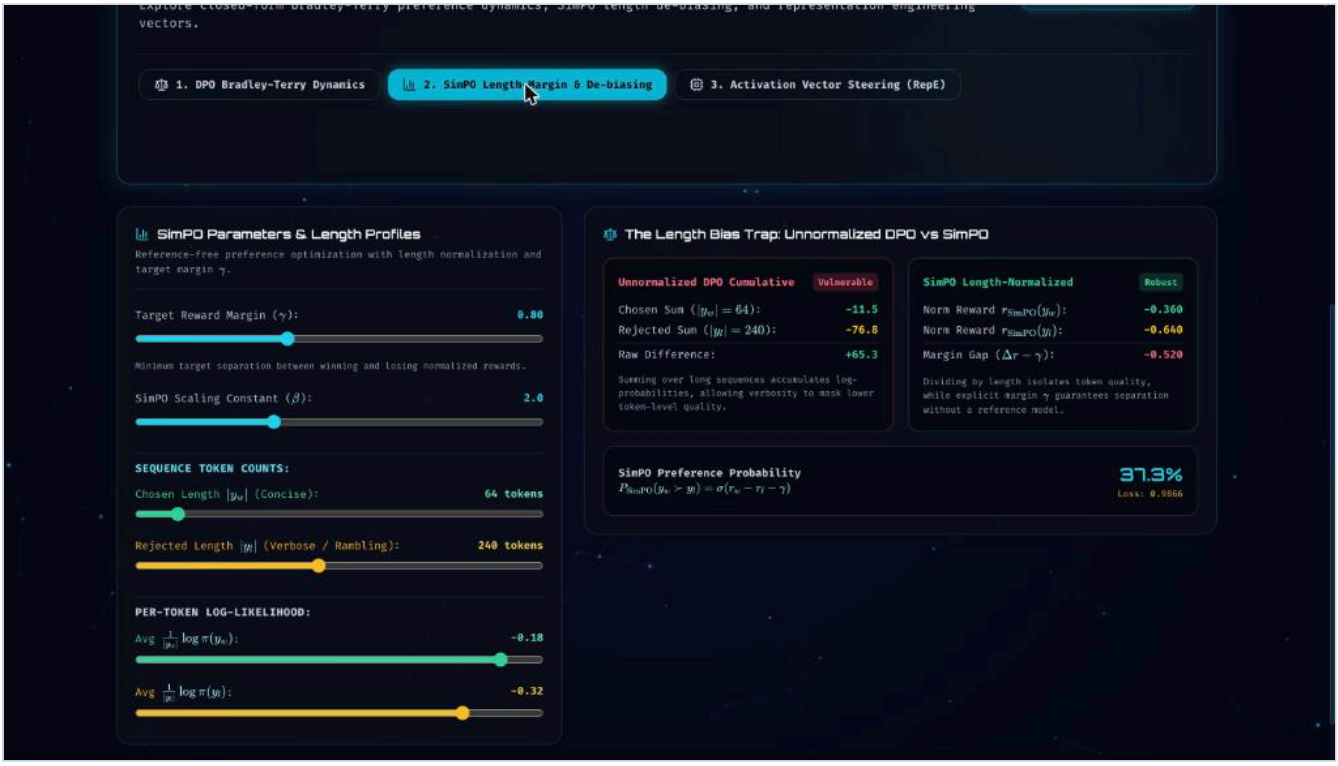
Screenshots



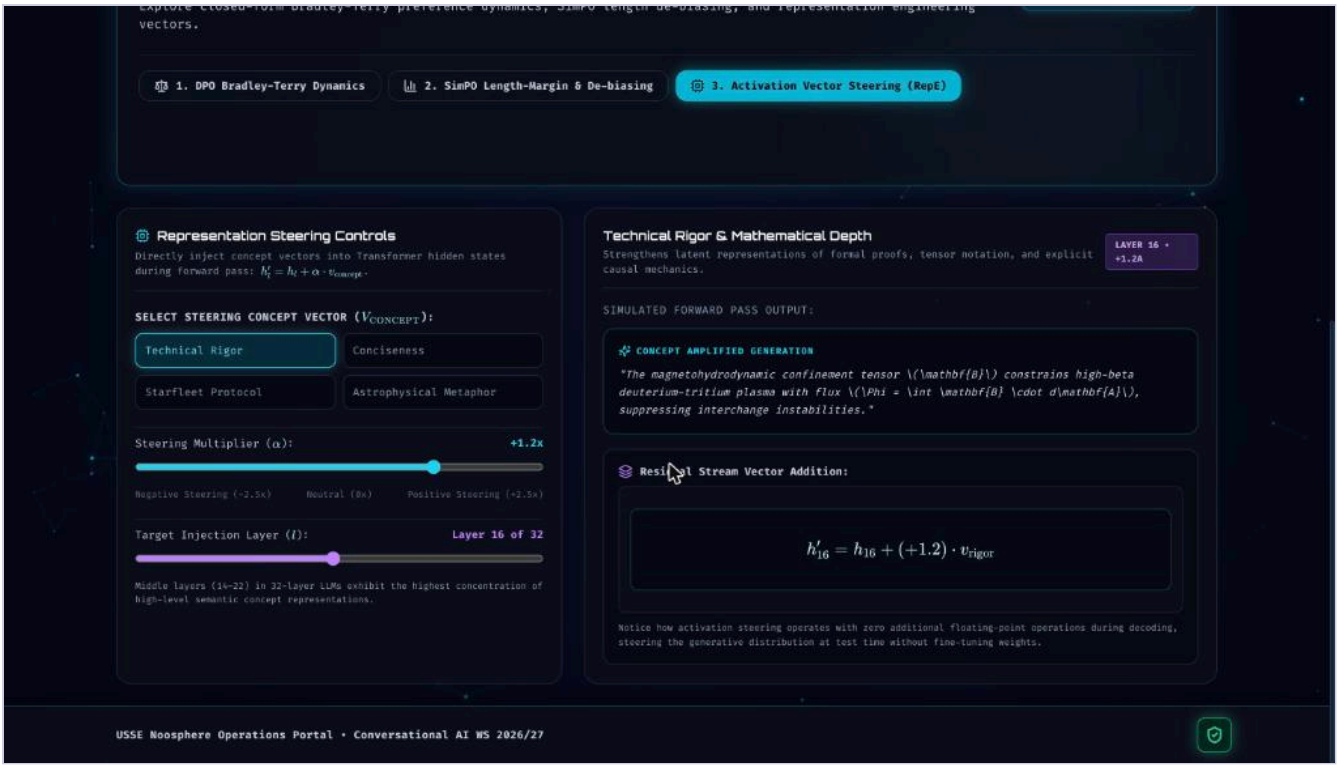
01_dpo_beta0.10_default.jpg



02_dpo_beta0.14.jpg



03_simpo_gamma0.80_beta2.0.jpg



04_steering_layer16of32_alpha1.2.jpg